

Important eqns:

S11 – Reflected Voltage

S21 – Transmitted Voltage

Z – Input impedance ($Z = Z_0(1 + S_{11})/(1 - S_{11})$)

Eqn. Flow -> S11 -> Z -> material response -> conductivity

Liftoff – Probe coil must be a certain distance away from the metal and not in contact. This can be done by placing a nonconductive material between the probe and metal. IN mms (0.5mm - 2mm maybe). The distance must be kept constant for consistent results.

Process flow:

1. Connect an AC power source to a coil that would be placed close to a conductive material for testing.
 - The coil could be an air-core pancake coil or PCB coil.
 - 50 Ohm coaxial cable for connection.
2. Once activated, a magnetic field is created from the AC signal creates a alternating voltage/current that produces a magnetic field. This magnetic field interacts with the metal object in testing, creating an eddy current inside of it. The eddy current then creates its own magnetic field trying to cancel out the coils magnetic field.
3. This changes the flux in the magnetic field which in turn changes the inductance of the coil. This ultimately will change the reactance which is what we will try and achieve through deriving S11.